iOS-BLE-SDK Documentation

Modification Record

Version Number	Modification time	Modifications
211.4	2023.4.7	First Edition
211.6	2023.5.10	Optimize related interfaces
211.8	2023.6.6	Add real-time data and blood sugar
212.4	2023.7.28	Add custom push message

1. Introduction

Purpose

Provide third-party companies with an iOS protocol SDK that can match smart watches. The SDK provides the main bracelets and mobile phones

The interaction protocol between them.

Note: The Bluetooth in the demo is the basic function and is only for reference. This SDK is only responsible for the protocol part.

2. API access instructions

1. Add the SDK package file and header file to the project.

2. Sending of the Agreement

1. For the sending of the protocol, refer to the STBlueToothSender.h file and the STBlueToothString.h file.

2. Protocol Description: The protocol is divided into commands without parameters and commands with parameters. Commands without parameters are passed in

Function String See STBlueToothString file. For commands with parameters, see STBlueToothSender file.

Parameters of response type need to be passed in.

3. Agreement acceptance.

STBlueToothData class is responsible for processing all data received by the app. The returned data is described in the file. It mainly returns a json string.

3. Receiving Agreement Instructions

1. Bluetooth callback uniformly uses monitoring notification Nof_Revice_Data_Key

ÿ :[[NSNotificationCenterdefaultCenter]addObserver:selfselector:@selector(nofReviceData:)

name: Nof_Revice_Data_Key object:nil];

-(void) nofReviceData:(NSNotification*)noti{if(noti&&noti.object){

NSDictionary *dict =noti.userInfo; // dict is the command code data returned by the SDK callback

NSDictionary *responseObject =noti.object; // responseObject is the content data returned by the SDK callback

2. The dict format is as follows //**{**

parameter	illustrate	Data Types
ST_ErrorType_Key	Error code (enumeration)	**NSNumber**
ST_RevType_Key	Command code (enumeration)	**NSNumber**

NSNumber*errorType=dict[ST_ErrorType_Key]; // revType is the result enumeration returned by the SDK callback

```
typedef enum:NSUInteger{
    RES_Correct=0x00,/*Command is correct*/
    RES_CmdError=0x01,/*Command code error*/
    RES_CrcError=0x02,/*Check code error*/
    RES_LenError=0x03,/*Data length error*/
    RES_Invalid=0x04,/*Invalid data*/
    RES_Time_Out=0x05,/*Data timeout*/
}ERROR_TYPE;
```

NSNumber*revType=dict[ST_RevType_Key]; // revType is the command code enumeration returned by the SDK callback

```
typedef enum :NSUInteger{
    REV_Device_Bind=0x81, //Bind device
    REV_Device_State=0x02, //Device status (get)
    SET_Device_State=0x82, //Device status (settings)
    REV_Device_Lost=0x03, //Find device
    REV_Device_Camera=0x04, //Photo control
    REV_Device_Battery=0x86, //Get the battery power
    REV_Device_Version=0x87, //Get device version information
    REV_Update_Time=0x08, //Time and time zone (get)
    SET_Update_Time=0x88, //Time and time zone (setting)
    REV_User_Info=0x09, //User information (get)
    SET_User_Info=0x89, //User information (settings)
    REV_Goal_Sport=0x0A, //One day's sports goal (get)
    SET_Goal_Sport=0x8A, //One day's sports goal (setting)
    REV_Device_Pair=0x8B, //Bluetooth system pairing
    REV_State_Notification=0x0C, //Message push switch (get)
    SET_State_Notification=0x8C, //Message push switch (settings)
    REV_Health_Current=0x8D, //Get the current device motion data
    REV_Health_Switch=0x0E, //Health data detection switch (get)
    SET_Health_Switch=0x8E, //Health data detection switch (setting)
    SET_Factory_Setting=0x8F, //Restore factory settings

    REV_Interval_HR=0x31, //heart rate detection interval (get)
    SET_Interval_HR=0xB1, //Heart rate detection interval (setting)
    REV_Contact_Common=0x32, //Common contacts (get)
    SET_Contact_Common=0xB2, // Frequent contacts (settings)
    REV_Contact_SOS=0x33, //Emergency contact (get)
    SET_Contact_SOS=0xB3, //Emergency contact (settings)
    REV_Remind_NoDisturb=0x34, //Do not disturb (get)
    SET_Remind_NoDisturb=0xB4, // Do not disturb (setting)
```

REV_Remind_AlarmClock=0x35, //Alarm clock (get)

SET_Remind_AlarmClock=0xB5, //Alarm clock (setting)

REV_Remind_Sedentary=0x36, //Sedentary reminder (get)

SET_Remind_Sedentary=0xB6, //Sedentary reminder (setting)

REV_Remind_DrinkWater=0x37, //Drink water reminder (get)

SET_Remind_DrinkWater=0xB7, //Drink water reminder (setting)

SET_Info_Weather=0xB9, //Push weather

REV_Alarm_Even=0x3B, //Event reminder (get)

SET_Alarm_Even=0xBB, //Event reminder (setting)

REV_History_Sport=0xE1, //Synchronous movement

REV_History_Step=0xE2, //synchronous step counting/sleep

REV_History_HR=0xE3, //Synchronize heart rate

REV_History_BP=0xE4, //Synchronize blood pressure

REV_History_BQ=0xE5, //Synchronize blood oxygen

REV_History_Pressure=0xE6, //Synchronous pressure

REV_History_Met=0xE7, //Synchronous Met

REV_History_Temp=0xE8, //Synchronize temperature

REV_History_ValidDate=0xE9, //Get the valid date list

REV_Bin_File_Info=0xEA, //Send file information

REV_Bin_File=0xEB, // file transfer

REV_Dial_Info=0xEC, //Get dial information

SET_Dial_Current=0xED, //Switch the current display dial

REV_History_Mai=0xEE, //Synchronize Mai

"#$%&'(#)*&$+,)-./!!!!!!0!!!12314!!!!!55Real-time data switch 6 set 7 !!!!%"8$

%&'(#)*&$+,)-./!!!!!0!!!129:4!!!!!55Real-time data switch 6 obtain 7 !!!!%"8$

%&'(#)*&$;<#<!!!!!!0!!!123:4!!!!!55Real-time data upload 6 obtain 7

REV_ShutDown_CMD=0x90, //Device shutdown

%"8$=)>-?@A$+BC'@!!!!!!0!!!129D4!!!!!55 Synchronous blood sugar

}REV_TYPE;

3. The format of responseObject is as follows //**{**

1. Receive binding

parameter	Indicates	Data Types
ST_DeviceBindKey	whether the binding is successful (1: success 0: failure)	**NSNumber**

2. Device status

parameter	illustrate	Data Types
ST_HourSystemKey	Time format setting (0: 24-hour system, 1: 12-hour format)	**NSNumber**
ST_UnitKey	Unit system selection (0: Metric, 1: Imperial)	**NSNumber**
ST_TemperatureUnitKey	Temperature system selection (0: Celsius, 1: Fahrenheit)	**NSNumber**
ST_LanguageDataKey	Watch language (0 Chinese Simplified, 1 Chinese Traditional, 2 English, 3 Russian, 4 French, 5 Spanish, 6 German, 7 Japanese, 8 Italian, 9 Korean, 10 Dutch, 11 Thai)	**NSNumber**
ST_BrightDurationKey	Backlight duration (seconds)	**NSNumber**
ST_BrightnessKey	Screen brightness (percentage)	**NSNumber**
ST_TrunWristKey	Lift to light up switch	**NSNumber**

3. Find the device

parameter	illustrate	Data Types
ST_FindDeviceKey	Check the phone (1: success 0: failure)	**NSNumber**

4. Photo control

parameter	illustrate	Data Types
ST_PhotoGraphModeKey	1: (Enter the photo taking interface) 2: (Exit the photo taking interface) 3: (Click on the app to take a photo)	**NSNumber**

5. Get battery power

parameter	illustrate	Data Types
ST_GetElectricityStateKey	Get charging status (1: charging) power	**NSNumber**
ST_GetElectricityKey	percentage	**NSNumber**

6. Get device information

parameter	illustrate	Data Types

ST_GetVersionNumberKey	Firmware version number	NSString
ST_GetUIVersionNumberKey	Firmware UI version number	NSString
ST_GetDeviceMtuKey	Device receiving buf size	**NSNumber**
ST_GetDeviceWidthKey	Device LCD Width	**NSNumber**
ST_GetDeviceHeightKey	Equipment LCD High	**NSNumber**
ST_GetDeviceShapeKey	Get device shape (0: round screen, 1: square screen)	**NSNumber**
ST_GetDeviceModelKey	device model batch number (string X01G001ÿ	NSString
ST_ForcedUpgradeUIKey	Is it necessary to force a UI upgrade?	**NSNumber**

7. Get the time zone

parameter	illustrate	Data Types
ST_YearKey	Year	NSNumber
ST_MonthKey	moon	NSNumber
ST_DayKey	day	NSNumber
ST_HourKey	hour	NSNumber
ST_MinuteKey	point	NSNumber
ST_SecondKey	Second	NSNumber
ST_ZoneKey	Time Zone	NSNumber

8. User Information

parameter	illustrate	Data Types
ST_UserGenderKey	Gender, 0: female, 1: male	NSNumber
ST_UserAgeKey	age	NSNumber
ST_UserHeightKey	Height (unit: 1CM) Weight	NSNumber
ST_UserWeightKey	(unit: 0.1KG)	NSNumber

9. Set a daily exercise goal

parameter	illustrate	Data Types
ST_TargetStepKey	Step Target	NSNumber
ST_TargetCalorieKey	Calorie goal (kcal)	NSNumber
ST_TargetDistanceKey	Distance goal (km)	NSNumber

10. Message push switch

parameter	illustrate	Data Types
ST_NotiAllKey	Main switch: 1 on 0 off	NSNumber
ST_NotiIncominglKey	Incoming call switch: 1 on 0 off	NSNumber
ST_NotiSmsKey	SMS switch: 1 on 0 off	NSNumber
ST_NotiEmailKey	mail switch: 1 on 0 off	NSNumber
ST_NotiTwitterKey	Twitter switch: 1 on 0 off	NSNumber
ST_NotiFacebookKey	Facebook switch: 1 on 0 off	NSNumber
ST_NotiWhatsappKey	WhatsApp switch: 1 On 0 Off	NSNumber
ST_NotiLineKey	Line switch: 1 on 0 off	NSNumber
ST_NotiSkypeKey	Skype switch: 1 on 0 off	NSNumber
ST_NotiQqKey	QQ switch: 1 on 0 off	NSNumber
ST_NotiWechatKey	WeChat switch: 1 on 0 off	NSNumber
ST_NotiInstagramKey	Instagram switch: 1 on 0 off	NSNumber
ST_NotiLinkedinKey	LinkedIn switch: 1 on 0 off	NSNumber
ST_NotiMessagerKey	Messenger switch: 1 on 0 off	NSNumber
ST_NotiVkKey	VK switch: 1 on 0 off	NSNumber
ST_NotiViberKey	Viber switch: 1 on 0 off	NSNumber
ST_NotiTelegramKey	Telegram switch: 1 on 0 off	NSNumber
ST_NotiKakaoTalkKey	KakaoTalk switch: 1 on 0 off	NSNumber
ST_NotiOtherKey	Other switches: 1 on 0 off	NSNumber

11. Get the current device display data

parameter	illustrate	Data Types
ST_GetCurrentValueStepKey	Total steps	NSNumber
ST_GetCurrentValueCalorieKey	Total calories (calories)	NSNumber
ST_GetCurrentValueDistanceKey	Total distance (m)	NSNumber
ST_GetCurrentValueSleepKey	Total sleep time (minutes) Total	NSNumber
ST_GetCurrentValueDeepSleepKey	deep sleep time (minutes) Total	NSNumber
ST_GetCurrentValueLightSleepKey	light sleep time (minutes) Heart	NSNumber
ST_GetCurrentValueHRKey	rate	NSNumber
ST_GetCurrentValueHeightBPKey	high pressure	NSNumber
ST_GetCurrentValueLowBPKey	Low pressure	NSNumber
ST_GetCurrentValueBOKey	Blood oxygen	NSNumber
ST_GetCurrentValueHPKey	pressure	NSNumber
ST_GetCurrentValueMETKey	Meto	NSNumber
ST_GetCurrentValueMAIKey	MIA	NSNumber
ST_GetCurrentValueTemperatureKey	Temperature (divided by 100 for display)	NSNumber

12. Get health data detection switch

parameter	illustrate	Data Types
ST_SwitchHRKey	Heart rate master switch: 1 on 0 off	NSNumber
ST_SwitchBPKey	Blood pressure master switch: 1 on 0 off	NSNumber
ST_SwitchBOKey	Blood oxygen master switch: 1 on 0	NSNumber
ST_SwitchPPKey	off Pressure master switch: 1 on 0 off	NSNumber
ST_SwitchTPKey	Temperature master switch: 1 on 0 off	NSNumber

13. Heart rate detection interval and range

parameter	illustrate	Data Types
ST_StartHourKey	Start Hour	NSNumber
ST_StartMinsKey	Start minute	NSNumber
ST_EndHourKey	End Hour	NSNumber
ST_EndMinsKey	End Minute	NSNumber
ST_IntervalKey	Detection cycle (in minutes)	NSNumber
ST_ThresholdKey	High static heart rate alarm threshold	NSNumber

14. Frequent/Emergency Contacts

parameter	illustrate	Data Types
ST_ContactsKey	Contact	NSArray
ST_ContactNameKey	Name	NSString
ST_ContactPhoneKey	Telephone	NSString

15. Do Not Disturb

parameter		Data Types
ST_StartHourKey	Description	NSNumber
ST_StartMinsKey	Start Hour Start Minute	NSNumber
ST_EndHourKey	End Hour	NSNumber
ST_EndMinsKey	End Minute	NSNumber
ST_NoDisturbOpenKey	Timer switch	NSNumber
ST_NoDisturbAllDayKey	Do not disturb all day	NSNumber

16. Alarm Clock

parameter	illustrate	Data Types
ST_ClocksKey	Alarm clock	NSArray
ST_ClockHourKey	hour	NSNumber
ST_ClockMinsKey	point	NSNumber
ST_ClockOpenKey	Switch	NSNumber
ST_ClockCycleKey	week repeat (6-0)	NSString
ST_ClockTypeKey	type	NSNumber

17. Sedentary/drinking water reminder

parameter	illustrate	Data Types
ST_StartHourKey	Start Hour	NSNumber
ST_StartMinsKey	Start minute	NSNumber
ST_EndHourKey	End Hour	NSNumber
ST_EndMinsKey	End Minute	NSNumber
ST_AlarmOpenKey	switch	NSNumber
ST_AlarmIntervalKey	Reminder interval (in minutes)	NSNumber

18. Synchronize motion data

parameter	illustrate	Data Types
ST_YearKey	Year	NSNumber
ST_MonthKey	moon	NSNumber
ST_DayKey	day	NSNumber
ST_HourKey		NSNumber
ST_MinuteKey		NSNumber
ST_SecondKey		NSNumber
ST_GetSportRecordDurationKey	Hours, minutes, seconds Exercise duration (s)	NSNumber
ST_GetSportRecordTypeKey	Type of exercise	NSNumber
ST_GetSportRecordStepKey	Steps	NSNumber
ST_GetSportRecordDistanceKey	Distance (m)	NSNumber
ST_GetSportRecordSpeedKey	Speed (m/s)	NSString
ST_GetSportRecordCaloriesKey	Calories	NSNumber
ST_GetSportRecordPaceKey	Pace	NSNumber
ST_GetSportRecordStrideKey	Cadence	NSNumber
ST_GetSportRecordHRLengthKey	Heart Rate Length	NSNumber

ST_GetSportRecordHRDetailsKey	Heart rate data	NSString

19. Synchronize step/calorie/distance/sleep data

parameter		Data Types
ST_GetRecordDateTimeKey	Description Year, month, day	NSString
ST_GetRecordIntervalKey	(yyyyMMdd) Historical sampling interval (mins)	NSNumber
ST_GetRecordValueDataKey	Historical data	NSArray
ST_GetRecordTypeStepOrSleepKey	Data type (1 step counting, 2 sleep)	NSNumber
ST_GetRecordValueStepKey	Step counting	NSNumber
ST_GetRecordValueCalorieKey	Calories(cal)	NSNumber
ST_GetRecordValueDistanceKey	Distance(m)	NSNumber
ST_GetRecordValueSleepKey Sleep (1 start to fall	asleep 2 light sleep 3 deep sleep 4 awake)	NSNumber

20. Synchronous heart rate/blood pressure/blood oxygen/pressure/METO/temperature/MAL/blood sugar data

parameter		Data Types
ST_GetRecordDateTimeKey	Description Year, Month, and Day (yyyyMMdd)	NSString
ST_GetRecordIntervalKey	Historical sampling interval (mins)	NSNumber
ST_GetRecordValueDataKey	Historical data: 1 minute per group (high and low pressure 2 as 1 group, Heart rate/blood pressure/blood oxygen/pressure 0 and 255 are abnormal numbers (need to be filtered)	NSString

21. Get a list of valid dates for data

parameter	illustrate	Data Types
ST_GetRecordValidDateKey	Valid date list year	NSArray
ST_YearKey		NSNumber
ST_MonthKey	month	NSNumber
ST_DayKey	day	NSNumber

22. Get dial information

parameter		Data Types
ST_GetDialInfoKey	Description dial information list	NSArray
ST_GetDialUsedKey	1 means the current display dial	NSNumber
ST_GetDialIdKey	Dial ID	NSNumber

ST_GetDialRGBKey	Color value (RGB565)	NSNumber
ST_GetDialModeKey	Position 1: 5 Top 2: Middle 3: Bottom	NSNumber

23. Get event reminder information

parameter	illustrate	Data Types
ST_EvensKey	Event reminder list	NSArray
ST_EvenYearKey	Year	NSNumber
ST_EvenMonthKey	moon	NSNumber
ST_EvenDayKey	day	NSNumber
ST_ClockHourKey	hour	NSNumber
ST_ClockMinsKey	point	NSNumber
ST_EvenTypeKey	Reminder type (1, 2, 3, 4)	NSNumber
ST_EvenCycleTypeKey	Repeat type (1: Single 2: Day 3: Week 4: Month 5: Year)	NSNumber
ST_EvenCycleValueKey	Weekly repeat value (bit 0 Monday)	NSString
ST_EvenContentKey	Reminder content	NSString

24. Real-time data switch

parameter	illustrate	Data Types
ST_GetRealTimeGsensorOnKey	gsensor	NSArray
ST_GetRealTimeSdcOnKey	Steps, calories, distance	NSNumber
ST_GetRealTimeHrOnKey	Heart rate,	NSNumber
ST_GetRealTimeBpOnKey	blood	NSNumber
ST_GetRealTimeBqOnKey	pressure, blood oxygen	NSNumber
ST_GetRealTimeTpOnKey	temperature	NSNumber
ST_GetRealTimeBsOnKey	blood sugar	NSNumber

4. Send agreement instructions

1. STBlueToothSender class

example:

Step 1: NSData*data =[STBlueToothSender writeDeviceDateTime];

Step 2: [STBleManager.sharedInstancewriteCommand:data];

The following is the details of sending data in the "first step"

1. Bind device

+(NSData*)writeDeviceBind;

2. Device status

+(NSData*)readDeviceState;

+(NSData*)writeDeviceState:(STDeviceState*)model;

STDeviceState Details

parameter	illustrate	Data Types
STITimeMode	Time format	enum
STUnit	unit	enum
STITemperatureUnit	Temperature Units	enum
STILanguage	Language Selection	enum
brightDuration	Screen on time (seconds)	NSInteger
brightness	Screen brightness (percentage) Turn	NSInteger
trunWrist	wrist to brighten screen (on/off)	BOOL

3. Find the device

+(NSData*)writeFindDevice;

4.Photo control

+(NSData*)writePhotoContrl:(STlPhotoContrl)contrl;

STlPhotoContrl Details

```
typedef enum :NSUInteger{
    ///Enter photo taking
    STlPhotoContrlStart=0x01,
    ///Exit photo taking
    STlPhotoContrlEnd=0x02,
    ///Click to take a photo
    STlPhotoContrlTakePhoto=0x03,

}STlPhotoContrl;
```

5. Get the battery level

+(NSData*)readDeviceBattery;

6. Get device version information

+(NSData*)readDeviceVersion;

7. Time and time zone

+(NSData*)readDeviceDateTime;

+(NSData*)writeDeviceDateTime;

8. User Information

+(NSData*)readUserInfo;

+(NSData*)writeUserInfo:(STlUserInfo*)model;

STlUserInfo Details

parameter	illustrate	Data Types
sex	Gender (default 0: female, 1: male)	NSInteger
age	Age (0x06~0x64, default 20) Height	NSInteger
height	(unit cm, default 170) Weight (unit	NSInteger
weight	0.1kg, default 600)	NSInteger

9. Daily exercise goals

+(NSData*)readSportGoal;

+(NSData*)writeSportGoal:(STlSportGoal*)model;

STlSportGoal Details

parameter	illustrate	Data Types
step	Step Target	NSInteger
calories	Calorie goal (unit: kcal) Distance	NSInteger
distance	goal (unit: km)	NSInteger

10.Bluetooth system pairing

+(NSData*)writeDevicePair;

11. Message push switch

+(NSData*)readMessageNotice;

+(NSData*)writeMessageNotice:(STMessageNotice*)model;

STMessageNotice Details

parameter	illustrate	Data Types

isAll	Main switch: 1 on 0 off	BOOL
isIncoming	Incoming call switch: 1 on 0 off	BOOL
isSms	SMS switch: 1 on 0 off	BOOL
isEmail	mail switch: 1 on 0 off	BOOL
isTwitter	Twitter switch: 1 on 0 off	BOOL
isFacebook	Facebook switch: 1 on 0 off	BOOL
isWhatsapp	WhatsApp switch: 1 On 0 Off	BOOL
isLine	Line switch: 1 on 0 off	BOOL
isSkype	Skype switch: 1 on 0 off	BOOL
isQq	QQ switch: 1 on 0 off	BOOL
isWechat	WeChat switch: 1 on 0 off	BOOL
isInstagram	Instagram switch: 1 on 0 off	BOOL
isLinkedin	LinkedIn switch: 1 on 0 off	BOOL
isMessager	Messenger switch: 1 on 0 off	BOOL
isVk	VK switch: 1 on 0 off	BOOL
isViber	Viber switch: 1 on 0 off	BOOL
isTelegram	Telegram switch: 1 on 0 off	BOOL
isKakaoTalk	KakaoTalk switch: 1 on 0 off	BOOL
isOther	Other switch: 1 on 0 off	BOOL

12. Get the current device display data

+(NSData*)readCurrentHealth;

13. Health data detection switch

+(NSData*)readHealthSwitch;

+(NSData*)writeHealthSwitch:(STHealthSwitch*)model;

STHealthSwitch Details

parameter	illustrate	Data Types
isHR	Heart rate master switch: 1 on 0 off	BOOL
isBP	Blood pressure master switch: 1 on 0 off	BOOL
isBO	Blood oxygen master switch: 1 on 0 off	BOOL
isPP	Pressure master switch: 1 on 0 off	BOOL
isTP	Temperature master switch: 1 on 0	BOOL
isBS	off Blood sugar master switch: 1 on 0 off	BOOL

14. Restore factory settings

+(NSData*)writeFactorySetting;

#pragmamark - **complex settings class**

16. Heart rate detection interval and range

+(NSData*)readConfigMeasureHR;

+(NSData*)writeConfigMeasureHR:(STConfigMeasureHR*)model;

STConfigMeasureHR Details

parameter	illustrate	Data Types
STIntervalTime	Time period	NSObject
interval	Detection cycle (in minutes)	NSInteger
thresholdHR	High static heart rate alarm threshold	NSInteger

STIntervalTime Details

parameter	illustrate	Data Types
startHour	Start Hour	NSInteger
startMinsv	Start minute	NSInteger
endHour	End Hour	NSInteger
endMins	End Minute	NSInteger

17. Frequently used contacts

+(NSData*)readCommonContacts;

+(NSData*)writeCommonContacts:(NSArray<STContacts*>*)modelArr;

STContacts Details

parameter	Indicate	Data Types
isSOS	whether it is an emergency	BOOL
name	contact name	NSString
phone	Telephone	NSString

18. Emergency Contacts

+(NSData*)readSosContacts;

+(NSData*)writeSosContacts:(NSArray<STContacts*>*)modelArr;

19. Do Not Disturb

+(NSData*)readNoDisturb;

+(NSData*)writeNoDisturb:(STNoDisturb*)model;

STNoDisturb Details

parameter	illustrate	Data Types
isOpen	Is it a timed do not disturb? Is	BOOL
isAllDay	it a do not disturb time period	BOOL
STIntervalTime	for the whole day?	NSObject

20. Alarm Clock

+(NSData*)readAlarmClocks;

+(NSData*)writeAlarmClocks:(NSArray<STAlarmClock*>*)modelArr;

STAlarmClock Details

parameter	illustrate	Data Types
isOpen	Alarm switch	BOOL
hour	Hour	NSInteger
minute	minute	NSInteger
STAlarmClockType	type	enum
cycle	Period (daily by default) false+(6-0)+(Every Sunday - Every Saturday))	NSArray

21. Sedentary reminder

+(NSData*)readSedentaryAlarmInterval;

+(NSData*)writeSedentaryAlarmInterval:(STAlarmInterval*)model;

STAlarmInterval Details

parameter	illustrate	Data Types
isOpen	switch	BOOL
STIntervalTime	Time period	NSObject
interval	Reminder interval (in minutes)	NSInteger

22. Drink water reminder

+(NSData*)readDrinkWaterAlarmInterval;

+(NSData*)writeDrinkWaterAlarmInterval:(STAlarmInterval*)model;

23. Push weather

+(NSData*)writeWeather:(NSArray<STWeather*>*)modelArr;

STWeather Details

parameter	illustrate	Data Types
temp	temperature	NSString
tempMin	Minimum temperature	NSString
tempMax	Maximum temperature	NSString
conditionCode	Weather status code (1, light rain, 2, moderate rain Rain, 3. Heavy rain, 4. Cloudy, 5. Overcast, 6. Sunny, 7. Haze, 8. Typhoon, 9. Thunderstorm, 10, hail 11. Light snow, 12. Moderate snow, 13. Heavy snow, 14. Rain and snow 15. Sandstorm, 16. School Add hail, 17, squall, 18, strong wind, 19. Light wind, 20. Tornado, 21. Heat With storm, 22, thunderstorm, 23, severe thunder Violence, 24, unknown ÿ	NSString
unit	Unit, 0 Celsius, 1 Fahrenheit	BOOL
windSpeed	Wind speed	NSString
humidity	Humidity (Relative humidity, percentage value)	NSString
still	Visibility, default unit: km	NSString
uvIndex	UV intensity	NSString
AQI	Air quality (1 excellent, 2 good, 3 poor)	NSString

24. Event Reminder

+(NSData*)readAlarmEvens;

#pragmamark- **data synchronization class**

25. Synchronous motion data (whether to resend 1: resend, 0: do not resend)

+(NSData*)readSportModeHistory:(BOOL)resend;

26. Synchronize step/calorie/distance/sleep data

///

///Get the historical record data for a certain day

///\paramdateStr date yyyyMMdd

///

+(NSData*)readStepAndSleepHistoryWithDate:(NSString*_Nonnull)dateStr;

27. Synchronize heart rate data

///

///Get the historical record data of a

certain day ///**\paramdateStr** date

yyyyMMdd ///

+(NSData*)readHeartRateHistoryWithDate:(NSString*_Nonnull)dateStr;

28.Synchronize blood pressure data

///

///Get the historical record data for a certain day

///\paramdateStr date yyyyMMdd

///

+(NSData*)readBloodPressureHistoryWithDate:(NSString*_Nonnull)dateStr;

29.Synchronize blood oxygen data

///

///Get the historical record data for a certain day

///\paramdateStr date yyyyMMdd

///

+(NSData*)readBloodOxygenHistoryWithDate:(NSString*_Nonnull)dateStr;

30.Synchronize pressure data

///

///Get the historical record data for a certain day

///\paramdateStr date yyyyMMdd

///

+(NSData*)readPhysicalPressureHistoryWithDate:(NSString*_Nonnull)dateStr;

31. Synchronize Metadata

///

///Get the historical record data of a

certain day ///**\paramdateStr** date yyyyMMdd

///

+(NSData*)readMetsHistoryWithDate:(NSString*_Nonnull)dateStr;

32.Synchronize temperature data

```
///
///Get the historical record data for a certain day
///\paramdateStr date yyyyMMdd
///
+(NSData*)readTemperatureHistoryWithDate:(NSString*_Nonnull)dateStr;
```

33.Synchronize Mai data

```
///
///Get the historical record data of a
certain day ///\paramdateStr date yyyyMMdd
///
+(NSData*)readMaiHistoryWithDate:(NSString*_Nonnull)dateStr;
```

34.Synchronize blood sugar data

```
///
///Get the historical data of a certain
day///\paramdateStr Date yyyyMMdd ///

+(NSData*)readSugarHistoryWithDate:(NSString*_Nonnull)dateStr;
```

35. Get the data validity date list (corresponding data type)

```
+(NSData*)readHistoryValidDate:(STHistoryCmd)cmd;
```

STHistoryCmd Details

```
typedef enum :NSUInteger{
    ST_History_Sport=0x61, //Synchronous movement
    ST_History_Step=0x62, //synchronous step counting/sleep
    ST_History_HR=0x63, //Synchronize heart rate
    ST_History_BP=0x64, //Synchronous blood pressure
    ST_History_BQ=0x65, //Synchronous blood oxygen
    ST_History_Pressure=0x66, //Synchronous pressure
    ST_History_Met=0x67, //Synchronous Met
    ST_History_Temp=0x68, //Synchronize temperature
    ST_History_Mai=0x6E, //Synchronize Mai
```

}STHistoryCmd;

#pragmamark **-dial settings class**

36. Get dial information

+(NSData*)readDeviceDialInfo;

37. Switch the current display dial

+(NSData*)writeCurrentDeviceDial:(int)Id;

Note, Id: watch face ID

#pragmamark- **UIImage to RGB565**

+(NSData*)getRGB656DataWith:(UIImage*)image;

//RGB888 to RGB565 array

+(NSData*)getRGB656DataWithR:(unsigned char)rG:(unsigned char)gB:(unsigned **char)b;**

#pragmamark-Real-time reporting

38. Real-time data switch (acquisition)

+(NSData*)readRealTimeSwitchs;

39. Real-time data switch (settings)

+(NSData*)writeRealTimeSwitchs:(STRealTimeSwitchs*)model;

STRealTimeSwitch Details

parameter	illustrate	Data Types
gsensor	gsensor real-time data	BOOL
sdc	Real-time data of steps, calories and distance	BOOL
hr	Real-time heart rate data	BOOL
bp	Real-time blood pressure data	BOOL
bq	Blood oxygen real-time data	BOOL
city	Temperature real-time data	BOOL
bs	Blood sugar real-time data	BOOL

40.Set shutdown (settings)

+(NSData*)writeShutDown;

2. STBlueToothData class

example:

Step 1: STBlueToothData*sender=[STBlueToothDatasharedInstance];

41. Write bin file data

- (void)writeBinFileInfo:(nonnullNSData *)info

Data : (nonnull NSData *) data

BlockReadInterval:(NSUInteger)blockReadInterval

Peripheral:(nonnull CBPeripheral*)peripheral

Characteristic:(nonnull CBCharacteristic*)characteristic

Progress:(ProgressBlock)progress;

Parameter Details

parameter		Data Types
info	Description	NSData
data	file brief file content	NSData
blockReadInterval	Maximum write length of the device	NSUInteger
peripheral	Device Object	CBPeripheral
characteristic	Device write characteristic value	CBCharacteristic
progress	Progress results written	ProgressBlock

//Event reminder (settings)

- (void)writeAlarmEvens:(NSArray<STAlarmEvent*>*)modelArr

BlockReadInterval:(NSUInteger)blockReadInterval

Peripheral:(nonnull CBPeripheral*)peripheral

Characteristic:(nonnull CBCharacteristic*)characteristic;

STAlarmEvent Details

parameter	illustrate	Data Types
year	Year	NSInteger
month	moon	NSInteger

day	day	NSInteger
hour	hour	NSInteger
minute	point	NSInteger
STAlarmEventType	type	enum
contentLen	Reminder content length	NSString
content	Reminder content (Unicode encoding, n maximum 254 bytes - 127 characters)	NSString
cycleType	Repeat type (1: Single 2: Day 3: Week 4: Month 5: Year)	NSInteger
cycle	Weekly repetition value (bit 0 Monday)	NSArray

42. Custom push messages (settings)

+(NSData *)writePushMessageTitle:(NSString *)titleMessage:(NSString *)message;

Parameter Details

parameter		Data Types
title		NSString
message	Description title content	NSString

43.Synchronize new sleep data

```
///
///Get the historical record data for a certain day
///\paramdateStr date yyyyMMdd
///
+(NSData*)readNewSleepHistoryWithDate:(NSString*_Nonnull)dateStr;
```

5. SDK Usage Principles

Note: When developers call the SDK to implement business logic, please strictly follow the order of [Bluetooth ready]->[Search device]->[Connect device]->[Device ready to read and write data]->[Function interface].

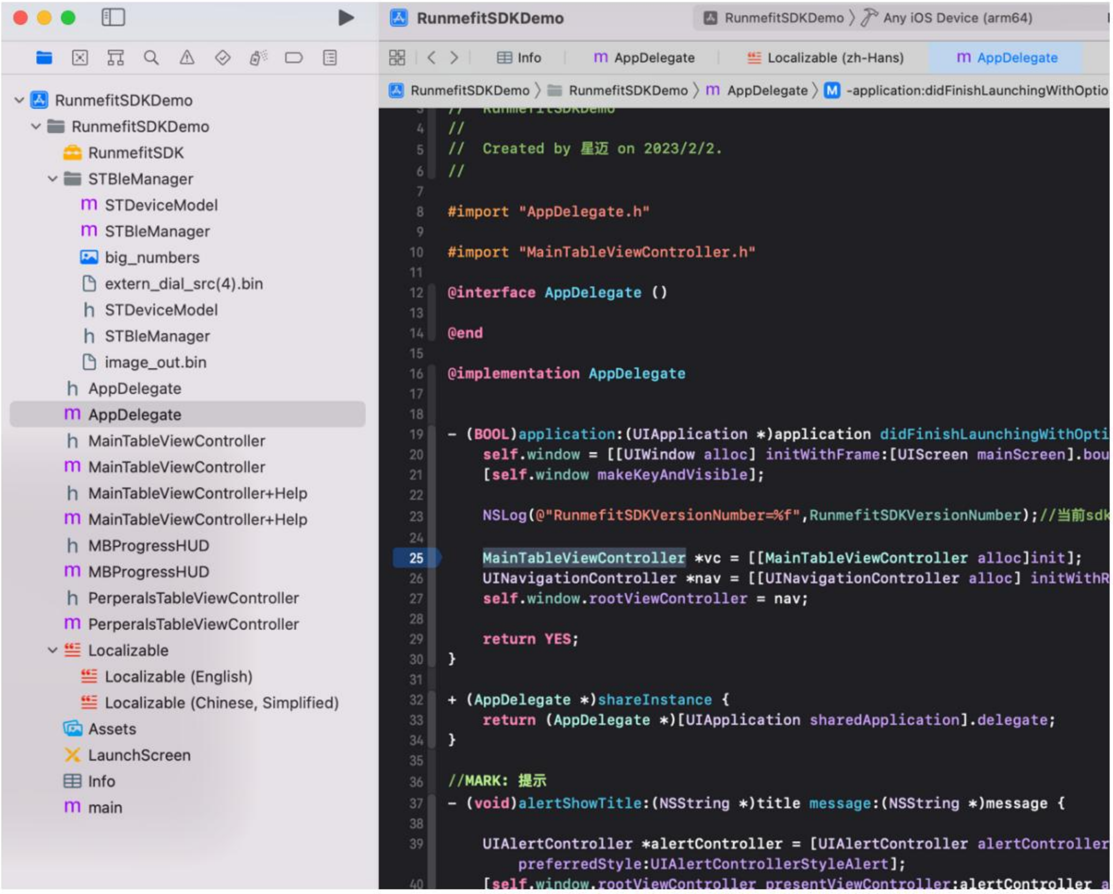
1. Scope of use

iOS-SDK Applicable to BLE4.0, iOS8.0 and above Xcode version 10.0 and above Supported languages

Objective-c, Swift4.0 and above

2. Manual integration of Xcode

Manually add RunmefitSDK.framework to your project directory. As shown in the figure



3. Equipment related operations

1. Initialization

The Bluetooth connection related classes are singletons. When needed, directly call the following method to obtain the object

STBleManager.sharedInstance

2. Start preparing for the search

a. Before searching for devices, make sure that Bluetooth is powered on. Please use STBleManager.sharedInstance setUpdateState callback method to determine Bluetooth status

@property(nonatomic,assign)BOOL stateOn; ///System Bluetooth switch

@property (nonatomic, strong)NSMutableArray<STDeviceModel*>*deviceModels; ///Search for devices

Start search: STBleManager.sharedInstancestartScan];

Stop searching: [STBleManager.sharedInstancestopScan];

b. Callback method to determine the search for Bluetooth devices

[STBleManager.sharedInstancesetUpdatePerpheral:^(NSArray<STDeviceModel*>*
_Nonnull deviceModels){

}];

STDeviceModel Details

parameter		Data Types
peripheral	Description	CBPeripheral
name	Device Object	NSString
mac	Device Name Device	NSString
rssi	Mac Address Device Signal	NSNumber

3. Connect your device

a. Pass the Bluetooth device model into the connection method

[STBleManager.sharedInstanceconnectPerpheral:deviceModel];

b. Callback method to determine the Bluetooth connection status

```
[STBleManager.sharedInstancesetUpdateConnect:^(BOOL connect){
        if (connect==YES){
            NSLog(@"%@",TRLocalizedString(@"Connection successful",nil));
        }else{
            NSLog(@"%@",TRLocalizedString(@"Connection failed",nil));
```

```
}
}];
```

4. Disconnect the device

a. Pass the Bluetooth device model to
the connection method [STBleManager.sharedInstancecancelPeripheral:deviceModel];

b. Callback method to determine the Bluetooth connection status

```
[STBleManager.sharedInstancesetUpdateConnect:^(BOOL connect){ if (connect==YES){
        NSLog(@"%@",TRLocalizedString(@"Connection successful",nil)); }
    else{
        NSLog(@"%@",TRLocalizedString(@"disconnected",nil));
    }
}];
```

5. Send instructions

```
[STBleManager.sharedInstancewriteCommand:data];
```